

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 12

TEMPERATURE
FACTORS CONTROLLING
TEMPERATURE (PART-2)



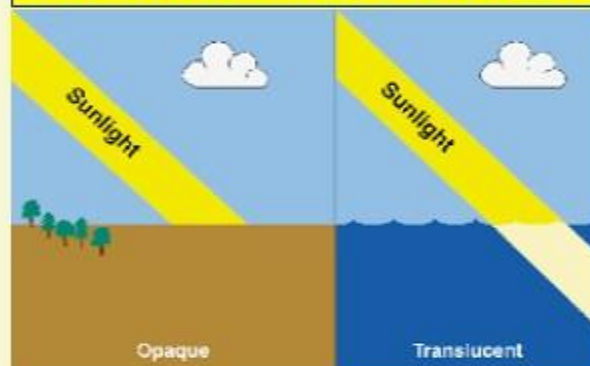
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



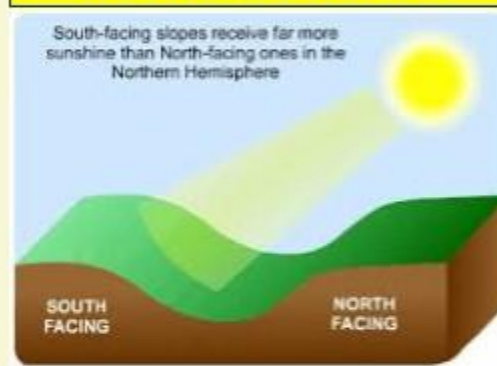
1. LATITUDES



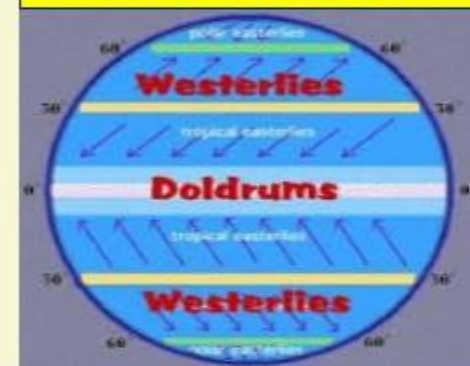
3. NATURE OF LAND & WATER



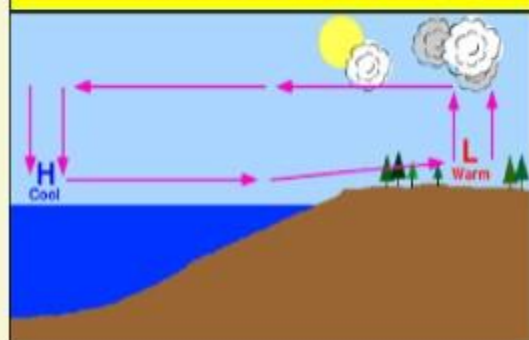
5. GROUND SLOPE



7. PREVAILING WINDS



2. DISTANCE FROM SEA



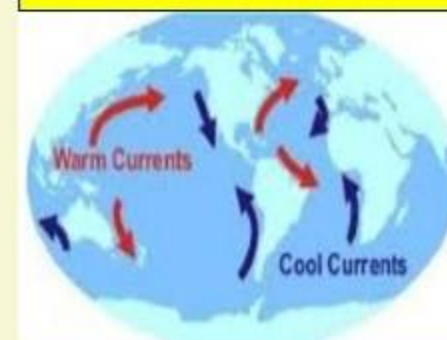
4. ALTITUDE



6. GROUND SURFACE



8. OCEAN CURRENTS



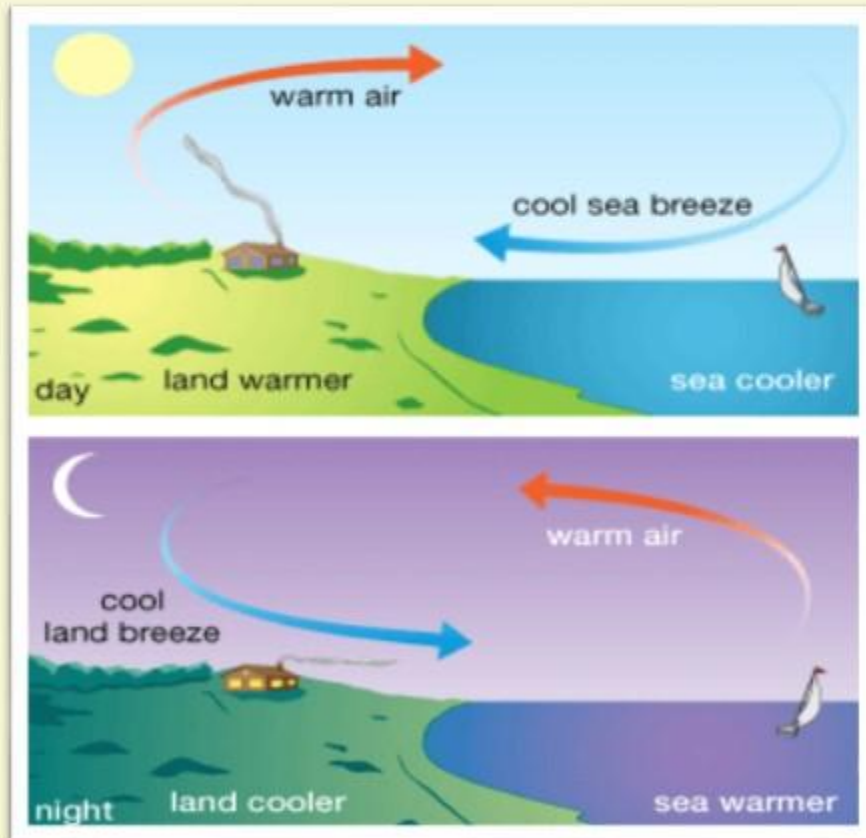


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 5

DISTANCE FROM THE SEA



- 1 The marine environment **moderates** the weather conditions of coastal areas.
- 2 There is more mixing of temperatures of the coastal seas and lands due to the daily **rhythm of land and sea breezes**.
- 3 Thus the **daily range of temperature** is minimum in coastal areas but it increases as the distance from coast increases.



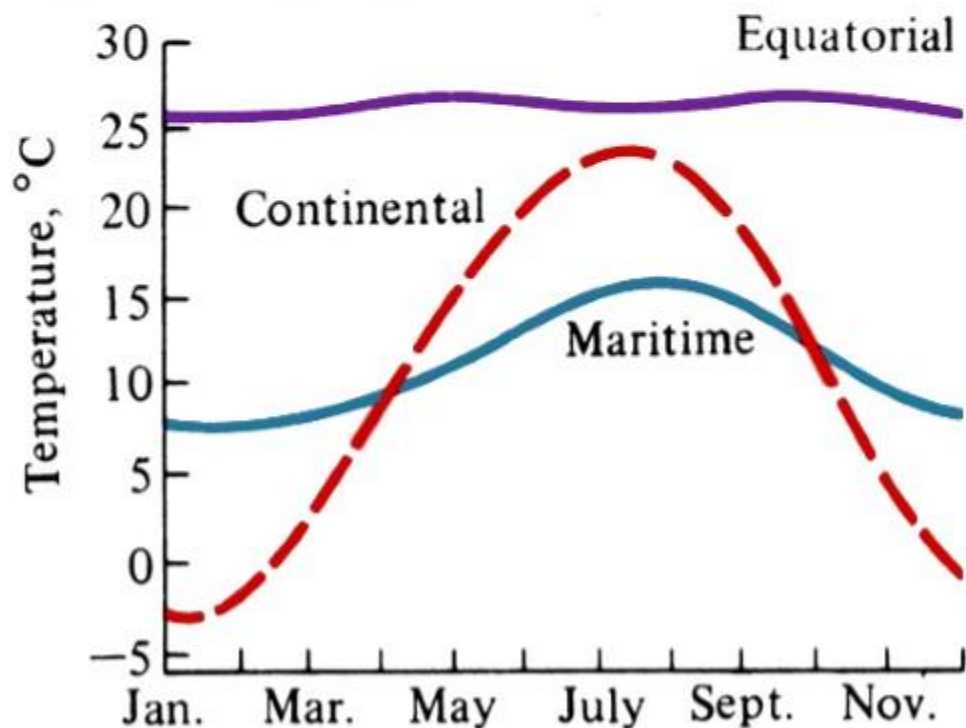
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 5

DISTANCE FROM THE SEA

ANNUAL RANGE OF TEMPERATURE



DIURNAL RANGE OF TEMPERATURE

- 1 The **diurnal temperature range (DTR)** is the difference between the daily maximum and minimum temperature
- 2 Minimum daily range of temperature is the characteristic feature of **MARINE CLIMATE**
- 3 Extremely high daily range of temperature characterizes **CONTINENTAL CLIMATE**

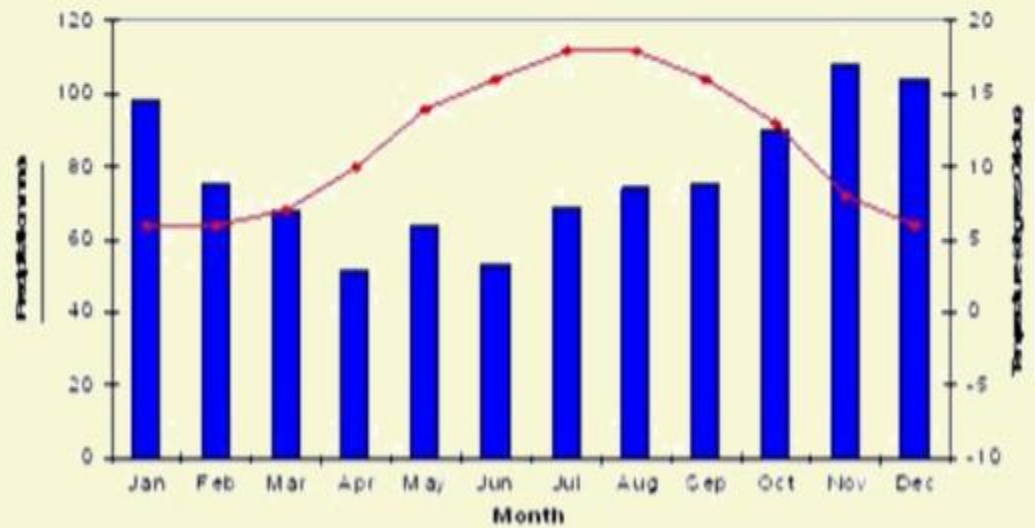
FACTOR : 5

DISTANCE FROM THE SEA

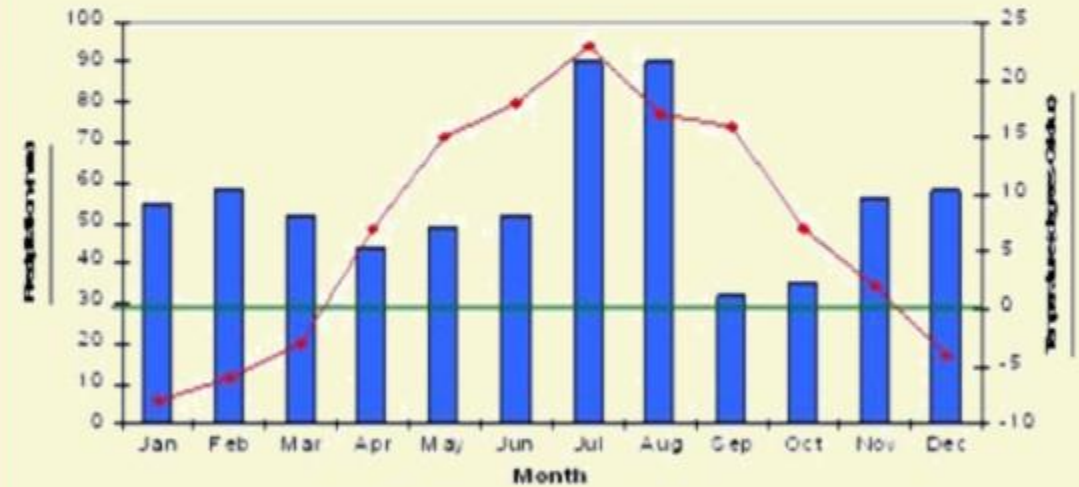
CASE STUDY OF PLYMOUTH AND KIEV



Average Climate in Plymouth
(Maritime)



Average Climate in Kiev,
Continental Interior





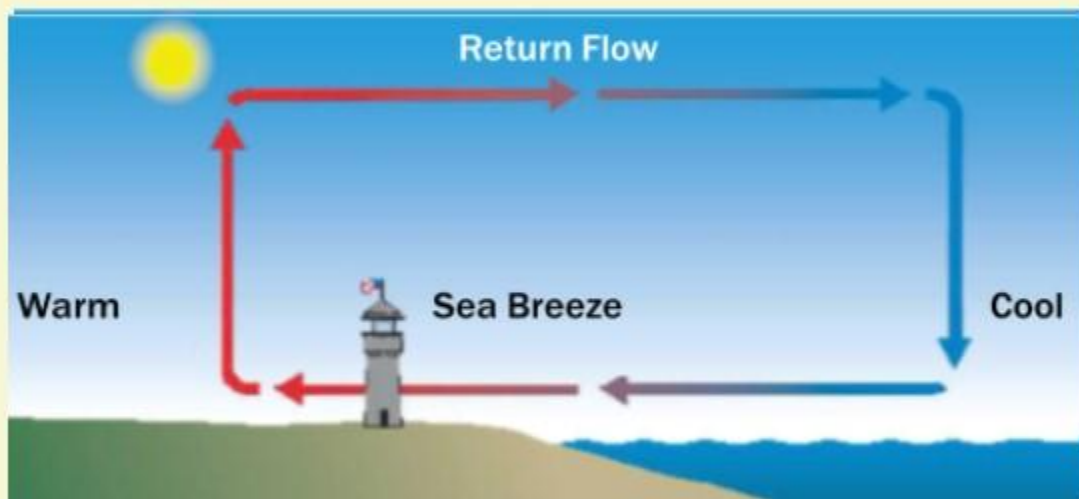
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 5

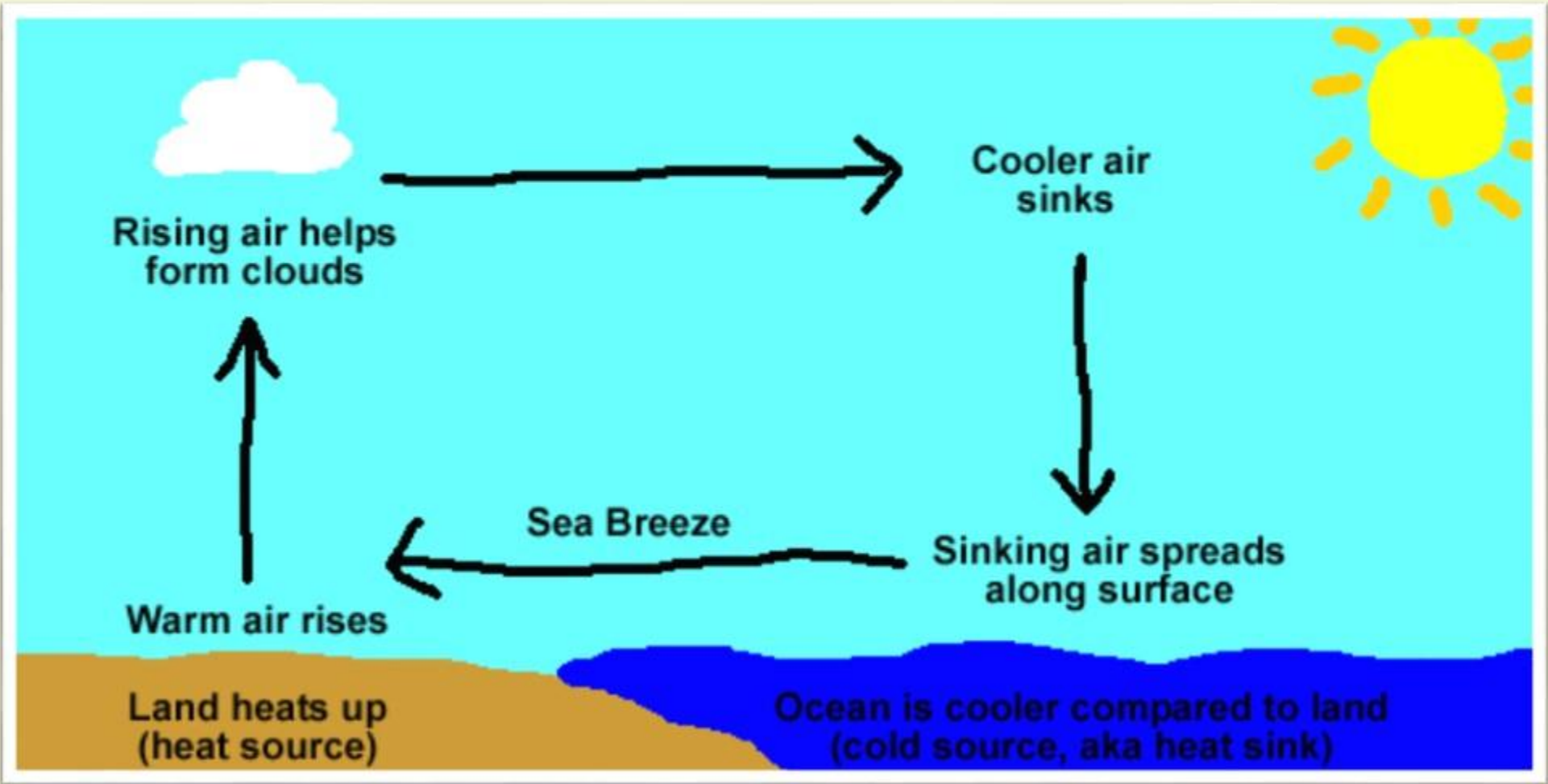
DISTANCE FROM THE SEA

LAND AND SEA BREEZE



- 1** In the day, the land heats up faster than the sea. The air above the land is heated, expands and rise.
- 2** Cool air above the sea is denser and moves in to replace the warmer air. This forms the sea breeze.
- 3** At night, the land cools faster than the sea.
- 4** As such, the air above the land is now cooler than the air above the sea
- 5** A convection current is set up in the opposite direction. This is called the land breeze.

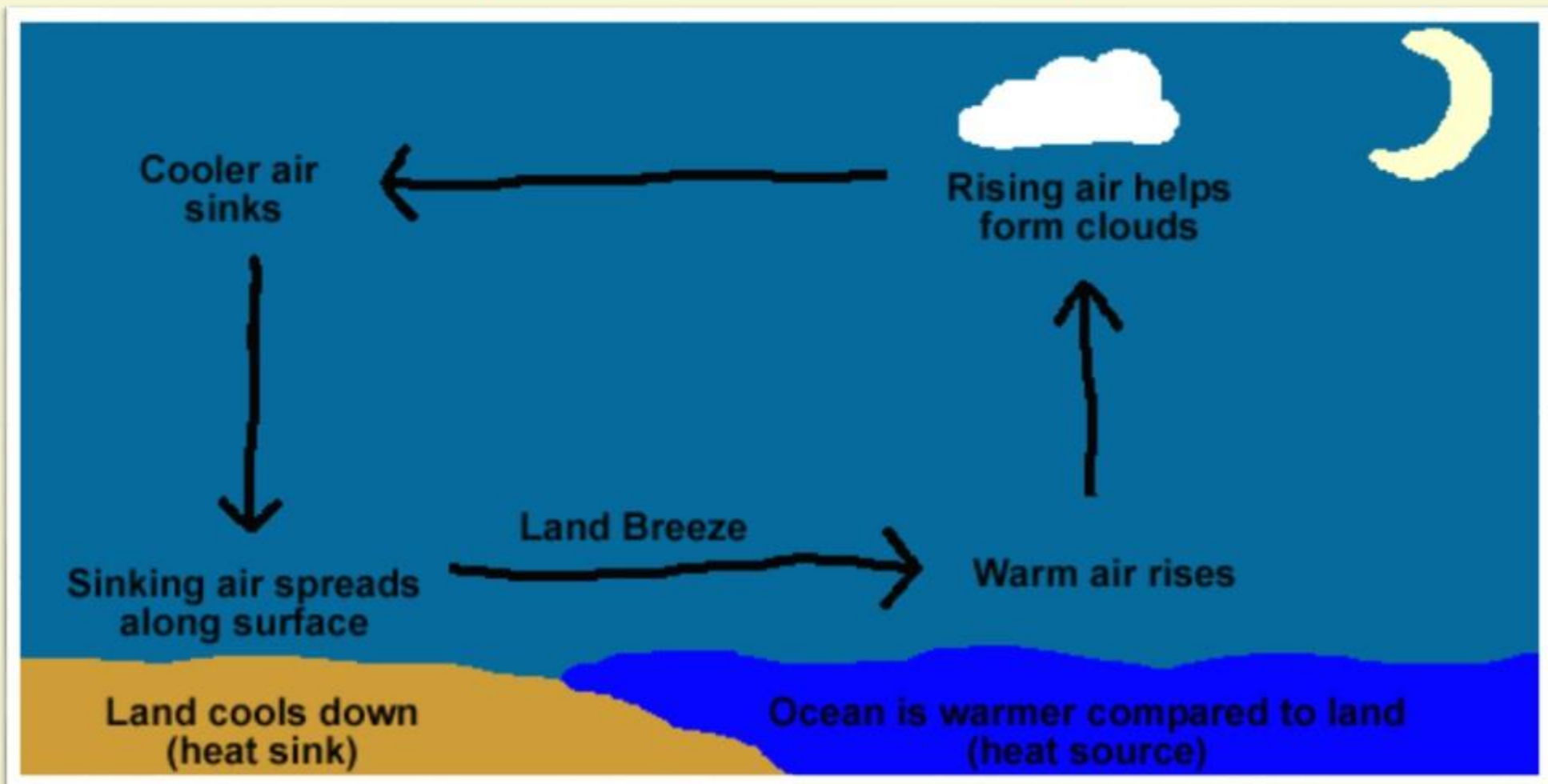
SEA BREEZE IN DAY



FACTOR : 5

DISTANCE FROM THE SEA

LAND BREEZE IN NIGHT



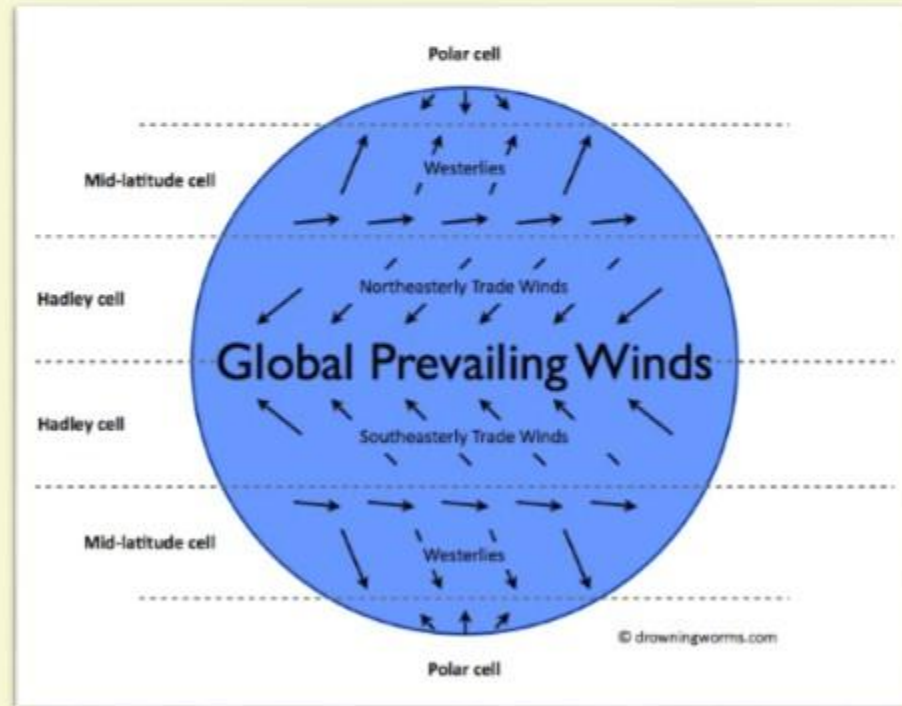


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS



The prevailing winds help in the redistribution of temperature and in carrying moderating effects of oceans to the adjacent coastal land areas.

- 1 Land and sea breezes
- 2 Mountain and valley breeze
- 3 Air masses

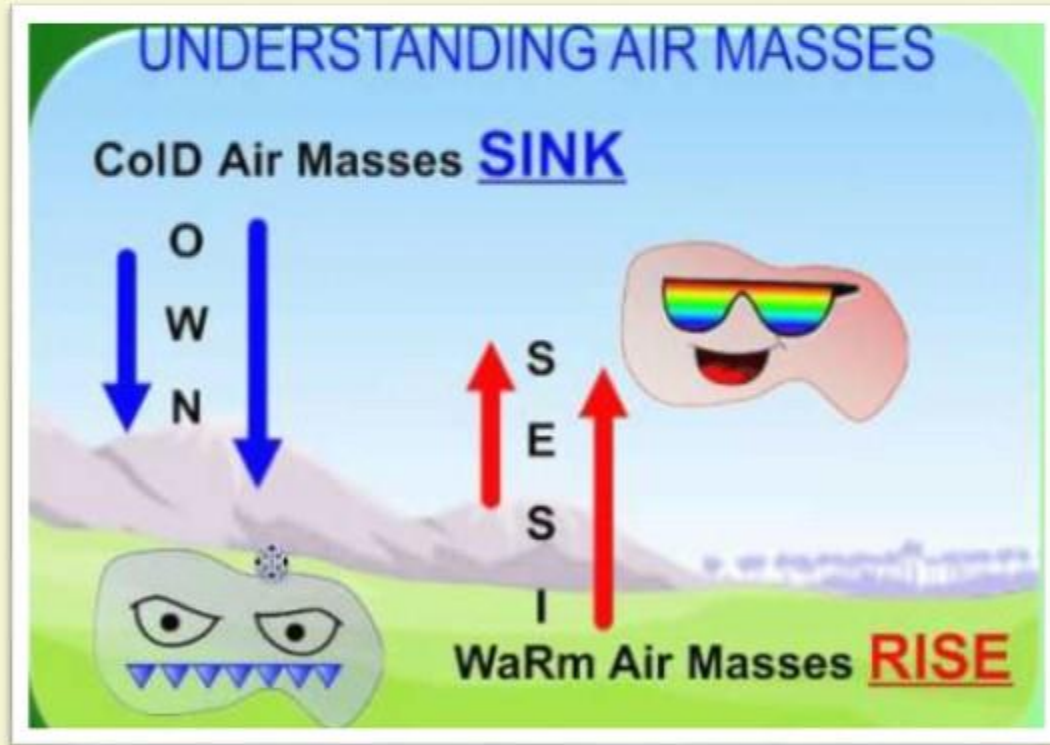


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS



WHAT IS AN AIR MASS?

- 1 Air masses are large body of air with similar temperature, pressure and humidity characteristics.
- 2 The places which come under the influence of warm air-masses experience higher temperature and the places that come under the influence of cold airmasses experience low temperature.



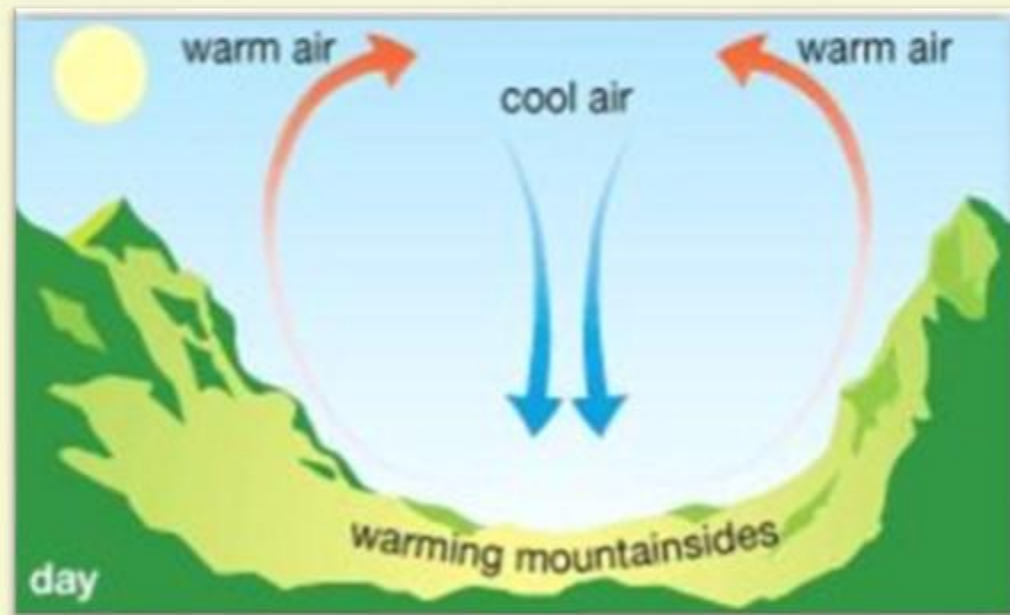
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS

VALLEY BREEZE



- 1 In the day, the mountain top heats up faster than valley creating a thermal low at mountain top.
- 2 Hence there would be an upslope movement of air from the valley towards the hill top known as the valley breeze.
- 3 Valley breezes reach mountain peaks and yield precipitation through cumulus clouds.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS

CUMULUS CLOUDS

Cumulus clouds are **puffy clouds** that sometimes look like **pieces of floating cotton**. These clouds grow upward, and they can develop into a **giant cumulonimbus**, which is a thunderstorm cloud.

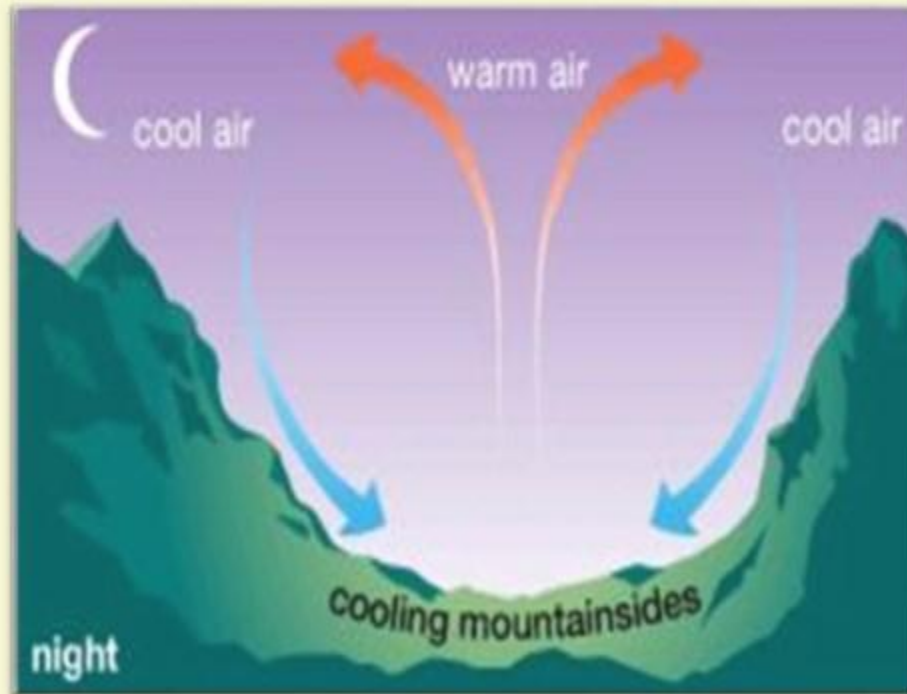


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 6

PREVAILING WINDS



MOUNTAIN BREEZE

- 1 In the night, the mountain top cools up faster creating a thermal high.
- 2 This creates a downslope movement of cool air from the hill top towards the valley known as the mountain breeze.

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